

Project Title: Machine Learning-Powered Scientific Computation

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Abstract

We propose a research program of integrating physical knowledge with powerful AI methods for expanding the scope of computer simulations for several challenging problems in the natural sciences. The careful design of these hybrid methods ensures the returned solutions can be used in downstream applications with confidence.

Keywords: AI for Science, Generative Model, Meta-Learning, Optimal Transport, Neural Network Architecture Design, Probabilistic Modeling.

Introduction

This grant will support several projects aiming at accelerating challenging simulation across various scientific domains using machine learning. Modern deep learning holds great promise for significantly advancing large-scale computations in science, but naïve drop-in of backbox off-the-shelf methods as replacement of the traditional pipeline typically witness limited success. We aim to exploit the problem structure, and bake in the right *inductive bias* to enable data-efficient algorithms that blend traditional numerical methods with powerful neural network models to accelerate these calculations with guarantees. The problems we aim to address span chemistry (transition-path sampling), biology (molecular dynamics), and physics (fluid flow simulation), and have significant impact from *drug discovery* to *climate prediction*. The research program pursued here will also span the whole range of theoretical work, methodology development, large-scale experimentation, and will be interdisciplinary in nature involving close collaborations with researchers across scientific fields.

Methods

We outline the approach for each project, along with the current state-of-the-art in detail below.

Project 1: Sampling Algorithms over Path Space

Transition Path Sampling (TPS) is a problem in computational chemistry where one is interested in sampling from high probability trajectories between two chemical states, A and B, separated by high energy barriers for a given chemical system (e.g., correspond to folded/unfolded states of a protein). Insights into the transition states of a chemical reaction have major implications in understanding disease and many material science applications. The goal of this project is to design algorithms to simulate time-series trajectories from a given posterior density over the path space -- the conditioning is on the final endpoint $x(T) = B$ in the case of TPS, but one can consider more general likelihood over the path $[x(0), \dots, x(T)]$.

In preliminary work [1], we put forth 2 algorithms based on controlled measure transport, and Wasserstein gradient flow, respectively, that can be integrated with neural network training for this challenging problem, which traditionally required significant compute and memory footprint, due to its rare-event nature. Both methods gradually trace out a curve of measures that converge to the desired posterior, and learn a “control” term parametrized by special NN

architecture (informed by our theoretical understanding of the problem structure) for tilting the current distribution towards the target. These methods offer significant speedups compared to classical MCMC-based local perturbative moves [2], since they are driven by an adaptive external control that accelerate the barrier crossing. The algorithms proposed in [1] work efficiently for a special case of the problem with a particular type of prior, which we would like to generalize to a broader class as follow-up work, along with larger scale experiments to test the robustness of the proposed methodology.

Project 2: Neural Partial Differential Equation Solver

Being a cornerstone of applied math and computer science, PDEs underpin modeling of complex dynamics across engineering disciplines, with applications in fluid dynamics, image processing and finance. This project investigates how PDE solvers can benefit from neural network training. More specifically, we study algorithms that have a learning component baked in. Instead of using generic hand-crafted update and discretization rules, we propose to use meta-learning, or learning to learn, to discover better algorithms. To give a concrete example, one can consider generalizing the simple space discretization $u'(x) = (u(x) - u(x + \delta_x))/\delta_x$ to $u'(x) = (a \times u(x) - b \times u(x + \delta_x))/\delta_x$ to allow for optimization of the (a,b) parameters using existing data, instead of fixed universally at (1,1).

We show in [3] that given access to data (solutions to similar PDEs previously solved using expensive traditional numerical methods), one can learn an improved *data-driven update* with higher fidelity (i.e., for a fixed stepsize, the discretization error is smaller). This trained NN-surrogate allows amortization and yields accurate solution at test time efficiently. Moreover, such family of methods can be trained end-to-end efficiently with the contemporary AutoDiff pipeline in deep learning, thanks to a dynamical system connection to the Neural ODE [4] and implicit layer architecture [5], where the gradient update needed for backpropagation can be computed cheaply using either adjoint method or implicit function theorem.

Distinct from Physics-Informed Neural Networks (PINNs) [6] and Neural Operators [7], these algorithms leverage significant problem structure and are built upon the insights from traditional PDE solvers with guarantees of consistency. With the main methodological framework laid out in the preliminary work [3], ongoing work involves conducting systematic numerical experiments on different classes of PDEs and benchmark the results against established solvers.

Project 3: Transferrable Generative Model for Molecular Dynamics Trajectories

This project concerns the design of efficient algorithms for conducting MD simulations, which are crucial for deeper understanding of many phenomena in biology and chemistry, with transformative implications for protein design and drug discovery. The development of this efficient “numerical emulator” will expand our capability to (1) move beyond the static single-structure prediction offered by AlphaFold [8] by capturing the probabilistic ensembles of conformations; (2) perform dynamical simulation that capture *how* important changes in proteins occur over longer time scale, as opposed to independent sampling from the equilibrium distribution only [9] that does not capture this “transient information”.

Using data collected from traditional compute-intensive MD methods, we leverage diffusion-based generative models to simulate high-probability molecular conformational changes over time. Ongoing work [10] leverages Sequential Monte Carlo and Reinforcement Learning to further improve the sample efficiency of this distribution learning task over the high-dimensional joint $\mathbb{R}^{d \times T}$ space, and also allows incorporation of any conditioning information / prior knowledge that a user may have at inference time seamlessly. These techniques, built on top of off-the-shelf diffusion models, allow the model to excel in a data-scarce regime without sacrificing the out-of-domain generalization capabilities across different biophysical systems.

Related Developments: Beyond these projects with specific applications in mind, the PI is also actively pursuing fundamental theoretical questions spanning optimal transport, MCMC sampling, dynamical systems and optimization in general metric space [11-13] – the connections among which lay the foundation for the algorithms proposed in Project 1-3 above. These studies will contribute to a precise understanding of the behavior of the algorithms under different parameter choices, and lead to practical insight into robustness and parameter tuning in software development.

Appendix A - Reference

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